

# SOCForge

## Building and Validating a Home Security Operations Center

*A Detection Engineering and Adversary Simulation Lab*

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Built Using

Wazuh • Sysmon • MITRE ATT&CK • Windows Endpoint Telemetry

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# 1. Executive Summary

Modern organizations rely on Security Operations Centers (SOCs) to monitor systems, detect threats, investigate suspicious activity, and respond to cybersecurity incidents. However, enterprise SOC environments often require expensive commercial tools and complex infrastructure. SOCForge was developed to demonstrate how a functional SOC environment can be designed and validated using open-source security technologies within a controlled home lab.

The objective of SOCForge was to build a simulated Security Operations Center capable of collecting endpoint telemetry, generating security alerts, analyzing attacker behavior, and supporting investigation workflows. The project combined multiple cybersecurity technologies, including Wazuh as the Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform, Sysmon for enhanced Windows endpoint visibility, and the MITRE ATT&CK framework for adversary behavior classification.

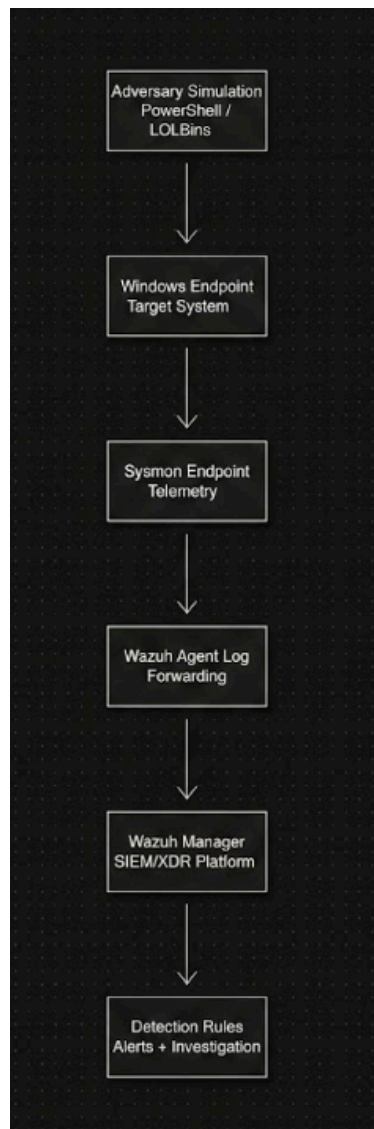
The lab environment was used to simulate realistic attacker activity and evaluate detection capabilities. Multiple attack techniques were tested, including PowerShell abuse, encoded command execution, download activity, and Living-off-the-Land Binary (LOLBIN) techniques using legitimate Windows utilities. These simulations allowed security events to be observed, analyzed, and mapped to relevant MITRE ATT&CK techniques.

SOCForge successfully demonstrated the complete security monitoring lifecycle, from endpoint activity generation and telemetry collection to alert review and investigation. The project highlighted the importance of endpoint visibility, effective detection engineering, and structured analysis when identifying malicious behavior.

This home SOC serves as a foundation for future improvements, including additional detection rules, automated response capabilities, threat intelligence integration, and expanded adversary simulations. The completed environment demonstrates how open-source tools can be combined to create a practical cybersecurity training platform that reflects real-world Security Operations Center processes.

## 2. Lab Architecture

The SOCForge environment was designed to replicate the core components and workflow of a Security Operations Center (SOC) within a controlled home lab. The architecture was built around endpoint visibility, centralized security monitoring, and detection analysis to demonstrate how security analysts collect and investigate suspicious activity.



*Figure 1. SOCForge Security Operations Center Architecture and Telemetry Flow*

Figure 1 illustrates the overall SOCForge architecture and the flow of endpoint telemetry from the monitored Windows system through Sysmon and the Wazuh platform. Security events are processed by custom detection rules, enabling analysts to investigate suspicious activity and classify observed techniques using the MITRE ATT&CK framework.

The SOCForge environment consisted of four primary components that worked together to provide endpoint visibility, centralized monitoring, and security analysis:

### **Windows Endpoint**

The Windows endpoint acted as the monitored system within the environment. This machine generated normal and simulated attacker activity, allowing security events to be collected and analyzed.

### **Sysmon Telemetry Collection**

Sysmon (System Monitor) was deployed on the Windows endpoint to provide enhanced visibility into endpoint activity. Compared to standard Windows event logging, Sysmon generates detailed telemetry on process creation, network connections, file activity, and other security-relevant events, providing the data required for threat detection and forensic investigation.



Export Formatted

792 available fields

Columns

Density

1 fields sorted

Full screen

|  | timestamp                   | agent.name | rule.description                                                                        |
|--|-----------------------------|------------|-----------------------------------------------------------------------------------------|
|  | Jul 13, 2026 @ 14:15:27.970 | Sarvesh    | Windows Logon Success                                                                   |
|  | Jul 13, 2026 @ 14:11:07.248 | Sarvesh    | A net.exe account discovery command was initiated                                       |
|  | Jul 13, 2026 @ 14:11:07.248 | Sarvesh    | A net.exe account discovery command was initiated                                       |
|  | Jul 13, 2026 @ 14:11:07.240 | Sarvesh    | Discovery activity executed                                                             |
|  | Jul 13, 2026 @ 14:11:07.240 | Sarvesh    | A net.exe account discovery command was initiated                                       |
|  | Jul 13, 2026 @ 14:11:07.235 | Sarvesh    | Discovery activity executed                                                             |
|  | Jul 13, 2026 @ 14:11:07.235 | Sarvesh    | Discovery activity executed                                                             |
|  | Jul 13, 2026 @ 14:11:07.227 | Sarvesh    | Discovery activity executed                                                             |
|  | Jul 13, 2026 @ 14:11:07.221 | Sarvesh    | Discovery activity executed                                                             |
|  | Jul 13, 2026 @ 14:11:01.889 | Sarvesh    | Powershell executed script from suspicious location                                     |
|  | Jul 13, 2026 @ 14:11:01.713 | Sarvesh    | Suspicious Windows cmd shell execution                                                  |
|  | Jul 13, 2026 @ 14:11:01.680 | Sarvesh    | Possible COM Hijacking evidence found in registry under HKCR\CLSID\{B11A45AF-1A63-4C    |
|  | Jul 13, 2026 @ 14:11:01.268 | Sarvesh    | Evidence of new service creation found in registry under HKLM\System\CurrentControlSet\ |
|  | Jul 13, 2026 @ 14:11:00.516 | Sarvesh    | Windows command prompt started by an abnormal process                                   |
|  | Jul 13, 2026 @ 14:11:00.288 | Sarvesh    | Suspicious Windows cmd shell execution                                                  |

*Figure 3. Wazuh Dashboard Displaying Centralized Security Monitoring*

Figure 3 demonstrates the centralized monitoring capabilities of the Wazuh platform. Security events collected from the Windows endpoint were processed and displayed within the dashboard, providing analysts with the information required to investigate suspicious behavior.

## Detection Rules and Analyst Workflow

Detection rules were used to identify suspicious behaviors associated with common attacker techniques. When activity matched a detection rule, alerts were generated and reviewed through an analyst workflow:

1. Alert identification
2. Event review and analysis
3. Evidence collection
4. MITRE ATT&CK technique classification
5. Investigation conclusion

This architecture allowed SOCForge to demonstrate how endpoint activity can be transformed into actionable security intelligence through centralized monitoring and detection engineering.

### **3. Environment Setup**

The SOCForge environment was built using open-source security technologies to create a functional Security Operations Center (SOC) capable of collecting endpoint telemetry, generating security alerts, and supporting incident investigations. The deployment process focused on establishing reliable communication between the monitored Windows endpoint and the centralized Wazuh platform.

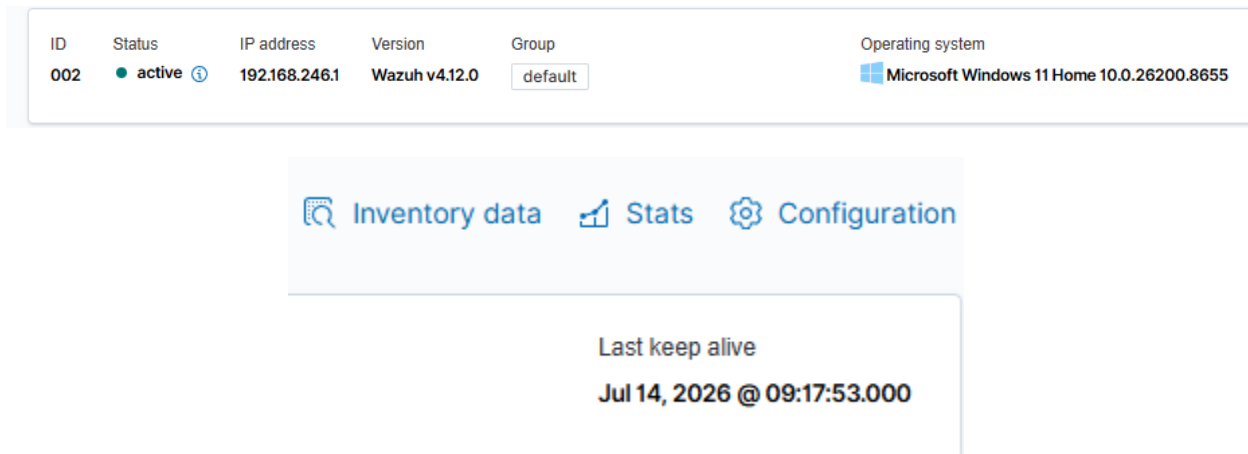
#### **Wazuh Deployment**

The Wazuh platform was deployed as the primary Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) solution. After installation, the web dashboard was configured to provide centralized visibility into endpoint activity, security alerts, and system health.

#### **Windows Agent Configuration**

A Wazuh agent was installed on the Windows endpoint to securely transmit telemetry to the Wazuh manager. After registration, communication between the endpoint and the management server was verified to ensure the agent was actively reporting system events.





*Figure 4. Windows Endpoint Successfully Registered with the Wazuh Manager*

Figure 4 confirms that the Windows endpoint was successfully registered with the Wazuh manager and was actively communicating with the centralized monitoring platform. Establishing this connection ensured that endpoint telemetry could be securely transmitted for analysis and detection.

## **Sysmon Deployment**

Microsoft Sysmon was deployed on the Windows endpoint using a customized configuration file designed to capture high-value security telemetry while reducing unnecessary log volume. Following installation, Sysmon successfully generated endpoint events that were forwarded to the Wazuh platform for centralized monitoring and analysis, confirming that the endpoint was producing the telemetry required for detection engineering and incident investigations.

SmartCard-TP

SmartScreen

SMBClient

SMBServer

SMBWitnessCl

StateRepositor

Storage-Tierin

StorageManag

StorageManag

StorageSetting

StorageSpaces

StorageSpaces

StorageSpaces

StorageSpaces

StorageSpaces

StorageVolum

StorDiag

Store

StorPort

Storsvc

Sudo

Sysmon

Operations

SystemSetting

TaskScheduler

TCPIP

TerminalServic

TerminalServic

TerminalServic

TerminalServic

TerminalServic

Operational

Number of events: 67,719 (!) New events available

| Level       | Date and Time        | Source | Event ID | Task C... |
|-------------|----------------------|--------|----------|-----------|
| Information | 7/14/2026 9:23:14 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:23:14 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:23:13 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:23:03 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:23:03 AM | Sysmon | 1        | Proces... |
| Information | 7/14/2026 9:22:59 AM | Sysmon | 1        | Proces... |
| Information | 7/14/2026 9:22:58 AM | Sysmon | 1        | Proces... |
| Information | 7/14/2026 9:22:58 AM | Sysmon | 1        | Proces... |
| Information | 7/14/2026 9:22:58 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:57 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:56 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:56 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:56 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:56 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:56 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:42 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:42 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:41 AM | Sysmon | 3        | Networ... |
| Information | 7/14/2026 9:22:41 AM | Sysmon | 3        | Networ... |

Figure 5. Sysmon Successfully Installed and Collecting Endpoint Events

Figure 5 demonstrates that Sysmon was successfully deployed and was actively collecting endpoint telemetry. The captured process creation and system events provided the detailed visibility required for Wazuh to generate alerts and support subsequent detection engineering activities.

## Telemetry Verification

Following deployment, endpoint telemetry was validated through the Wazuh dashboard to confirm that Sysmon events were successfully collected and processed. Active agent status, incoming security events, and endpoint logs verified that communication between all components was functioning correctly.

With the environment fully configured and endpoint telemetry successfully validated, the SOCForge platform was ready for the development and testing of custom detections. The

following section focuses on the detection engineering process and the validation of security rules against simulated adversary behavior.

## **4. Detection Engineering**

Detection engineering is the process of designing, implementing, and validating security detections that identify malicious or suspicious behavior within an environment. In SOCForge, detection engineering focused on developing and testing detection logic capable of identifying common attacker techniques using endpoint telemetry collected by Sysmon and analyzed through the Wazuh platform.

Each detection was evaluated by simulating adversary activity on the Windows endpoint and verifying that the corresponding events were successfully collected, processed, and classified. The detections were mapped to the MITRE ATT&CK framework to align observed behaviors with widely recognized adversary techniques and to provide standardized context for security investigations.

### **PowerShell Abuse Detection**

#### **Objective**

The first detection was designed to identify suspicious PowerShell execution commonly associated with attacker activity. Because PowerShell is frequently abused for post-exploitation, malware execution, and system reconnaissance, monitoring its execution is an important component of endpoint detection.

#### **Technique Being Detected**

The detection monitored PowerShell process creation events generated by Sysmon and analyzed through the Wazuh platform. Detection rules were configured to identify suspicious PowerShell execution patterns and generate alerts for analyst review.

## MITRE ATT&CK Mapping

- T1059.001 – Command and Scripting Interpreter: PowerShell

## Validation Results

Simulated PowerShell activity successfully generated security events that were collected by Sysmon and processed by Wazuh. The corresponding detection rule generated alerts, demonstrating that the SOC environment was capable of identifying suspicious PowerShell execution.

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t data.win.system.systemTime | 2026-07-10T18:39:59.2548042Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| t data.win.system.task       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| t data.win.system.threadID   | 95836                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| t data.win.system.version    | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| t decoder.name               | windows_eventchannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| t full_log                   | return { } } 'false'\\\\"currentDirectory": "C:\\\\Users\\sarvesh\\AppData\\Local\\Dawn\\", "user": "Sarvesh\\sarvesh", "logonGuid": "{5bad5dcd-fa72-6a36-f007-170000000000}", "logonId": "0x1707f0", "terminalSessionId": "1", "integrityLevel": "Medium", "hashes": "MD5=A97E6573B97B44C96122BFA543A82EA1, SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5F56381870317E8C46, IMPHASH=AFACF6DC9041114B198160AAB4D0AE77", "parentProcessGuid": "{5bad5dcd-3c7d-6a51-d7b4-090000000700}", "parentProcessId": "110632", "parentImage": "C:\\\\Users\\sarvesh\\AppData\\Local\\Dawn\\Dawn (Feather)\\Dawn (Feather).exe" "parentComm |
| t id                         | 1783708799.6312681                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| t input.type                 | log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| t location                   | EventChannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| t manager.name               | soc-lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| t rule.description           | Sysmon: Suspicious PowerShell execution detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| # rule.firedtimes            | 22                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| t rule.groups                | local, sysmon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| t rule.id                    | 100002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| # rule.level                 | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| rule.mail                    | true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| t rule.mitre.id              | T1059.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| t rule.mitre.tactic          | Execution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| t rule.mitre.technique       | PowerShell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| timestamp                    | Jul 10, 2026 @ 14:39:59.985                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Figure 6a. PowerShell Detection Alert

Figure 6a shows the security alert generated after the simulated PowerShell activity was executed. The alert confirmed that the custom detection successfully identified suspicious PowerShell execution within the monitored environment.

```
<rule id="100002" level="12">
  <if_sid>61603</if_sid>
  <field name="win.eventdata.image">powershell.exe</field>
  <field name="win.eventdata.commandLine">ExecutionPolicy Bypass</field>
  <description>Sysmon: Suspicious PowerShell execution detected</description>
  <mitre>
    <id>T1059.001</id>
  </mitre>
</rule>

</group>
```

*Figure 6b. Custom Detection Rule for Suspicious PowerShell Execution*

Figure 6b presents the custom Wazuh detection rule responsible for identifying suspicious PowerShell execution. The rule demonstrates how endpoint telemetry can be transformed into actionable security detections through detection engineering.

## Download Activity Detection

### Objective

The second detection focused on identifying suspicious download activity commonly associated with malware delivery and payload retrieval.

### Technique Being Detected

Detection rules monitored endpoint activity for behaviors consistent with downloading external files using PowerShell-based download methods.

### MITRE ATT&CK Mapping

- T1105 – Ingress Tool Transfer

## Validation Results

The simulated download activity generated endpoint telemetry that was successfully collected and processed by Wazuh. Detection rules produced alerts that enabled the activity to be reviewed and correlated during the investigation process.

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data.win.system.systemTime | 2026-07-11T18:22:26.6461827Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| data.win.system.task       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| data.win.system.threadID   | 95836                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| data.win.system.version    | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| decoder.name               | windows_eventchannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| full_log                   | r": "Sarvesh\\\\\\sarve", "logonGuid": "{5bad5dcd-fa/2-6a36-930/-1/000000000000}", "logonId": "0x1 / 0793", "terminalSessionId": "1", "integrityLevel": "High", "hashes": "MD5=A97E6573B97B44C96122BF A543A82EA1, SHA256=0FF6F2C948C7E2833A5F7E16DE1622E5DBA70396F31C7D5F56381870317E8C46, IMPHASH =AFACF6DC9041114B198160AAB4D0AE77", "parentProcessGuid": "{5bad5dcd-89d8-6a52-e4da-090000000 700}", "parentProcessId": "113448", "parentImage": "C:\\\\Windows\\\\\\System32\\\\\\WindowsPowerS hell\\\\\\v1.0\\\\\\powershell.exe", "parentCommandLine": "C:\\\\Windows\\\\\\System32\\\\\\Wind owsPowerShell\\\\\\v1.0\\\\\\powershell.exe\\\\\\", "parentUser": "Sarvesh\\\\\\sarve"}} |
| id                         | 1783794146.2492618                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| input.type                 | log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| location                   | EventChannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| manager.name               | soc-lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| rule.description           | Sysmon: PowerShell download activity detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| # rule.firedtimes          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| rule.groups                | local, sysmon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| rule.id                    | 100005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| # rule.level               | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| rule.mail                  | true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| rule.mitre.id              | T1059.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| rule.mitre.tactic          | Execution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| rule.mitre.technique       | PowerShell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| timestamp                  | Jul 11, 2026 @ 14:22:26.991                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Figure 7a. Wazuh Alert Detecting PowerShell Download Activity

Figure 7a illustrates the alert generated when PowerShell download activity was simulated. The successful detection confirmed that the environment could identify behaviors associated with payload retrieval and external file downloads.

```
<rule id="100005" level="12">
  <if_sid>61603</if_sid>
  <field name="win.eventdata.image">powershell.exe</field>
  <field name="win.eventdata.commandLine">Invoke-WebRequest</field>
  <description>Sysmon: PowerShell download activity detected</description>
  <mitre>
    <id>T1059.001</id>
  </mitre>
</rule>
```

*Figure 7b. Custom Detection Rule for PowerShell Download Activity*

Figure 7b displays the custom detection rule used to identify PowerShell download activity. This rule was designed to detect behaviors commonly associated with attacker attempts to retrieve malicious payloads.

## Living-off-the-Land (LOLBIN) Detection

### Objective

The final detection focused on identifying abuse of legitimate Windows utilities commonly leveraged by attackers to execute malicious activity while avoiding traditional malware detection.

### Technique Being Detected

The lab simulated the use of common LOLBINS, including Certutil, Mshta, and Rundll32. Detection rules monitored the execution of these utilities and generated alerts when suspicious behavior was observed.

### MITRE ATT&CK Mapping

- T1105 – Ingress Tool Transfer (Certutil)
- T1218.005 – Mshta
- T1218.011 – Rundll32

## Validation Results

Each simulated LOLBIN technique generated security events that were successfully captured by Sysmon and analyzed within Wazuh. Detection rules accurately identified the execution of these utilities, allowing the activity to be investigated and mapped to the appropriate MITRE ATT&CK techniques.

| timestamp                   | agent.name | rule.description                             | rule.level | rule.id |
|-----------------------------|------------|----------------------------------------------|------------|---------|
| Jul 11, 2026 @ 14:24:31.519 | Sarvesh    | Sysmon: Rundll32 execution detected (LOLBin) | 12         | 100008  |
| Jul 11, 2026 @ 14:24:26.493 | Sarvesh    | Sysmon: Mshta execution detected (LOLBin)    | 12         | 100007  |
| Jul 11, 2026 @ 14:24:22.165 | Sarvesh    | Sysmon: Certutil execution detected (LOLBin) | 12         | 100006  |

Figure 8a. Wazuh Events Detecting Living-off-the-Land Binary Execution

Figure 8a demonstrates the successful detection of LOLBIN execution within the Wazuh platform. Alerts generated during the simulation provided analysts with the information necessary to investigate legitimate Windows utilities being used for malicious purposes.

```
<rule id="100006" level="12">
  <if_sid>61603</if_sid>
  <field name="win.eventdata.image">certutil.exe</field>
  <description>Sysmon: Certutil execution detected (LOLBin)</description>
  <mitre>
    <id>T1105</id>
  </mitre>
</rule>

<rule id="100007" level="12">
  <if_sid>61603</if_sid>
  <field name="win.eventdata.image">mshta.exe</field>
  <description>Sysmon: Mshta execution detected (LOLBin)</description>
  <mitre>
    <id>T1218.005</id>
  </mitre>
</rule>

<rule id="100008" level="12">
  <if_sid>61603</if_sid>
  <field name="win.eventdata.image">rundll32.exe</field>
  <description>Sysmon: Rundll32 execution detected (LOLBin)</description>
  <mitre>
    <id>T1218.011</id>
  </mitre>
</rule>
</group>
```

Figure 8b. Custom Detection Rules for Living-off-the-Land Binary Techniques



Figure 8b presents the custom detection rules developed to identify the execution of common LOLBIN techniques, including Certutil, Mshta, and Rundll32. These rules expanded the SOC's ability to detect attacker behavior beyond traditional malware signatures.

## **5. Adversary Simulation**

Adversary simulation was performed to validate the effectiveness of the SOCForge detection capabilities under realistic attack scenarios. Rather than analyzing isolated events, this phase evaluated the complete attack workflow by executing common attacker techniques and confirming that endpoint telemetry, detection rules, and security alerts functioned together as expected. Each simulation was conducted within the controlled lab environment to ensure that malicious behavior could be safely reproduced and analyzed.

### **Phase A: PowerShell Attack Chain**

#### **Objective**

The first simulation focused on validating the detection of suspicious PowerShell activity commonly observed during the early stages of an attack. Multiple PowerShell techniques were executed to confirm that endpoint telemetry was successfully captured and that the corresponding Wazuh detection rules generated security alerts.

#### **Attack Sequence**

The simulated attack included the following behaviors:

- PowerShell execution
- Execution policy bypass
- Hidden PowerShell execution

- PowerShell download activity using Invoke-WebRequest

Each stage generated Sysmon process creation events that were forwarded to the Wazuh platform for analysis. The custom detection rules successfully identified the suspicious behavior, allowing the activity to be reviewed through the Threat Hunting interface and correlated with the appropriate MITRE ATT&CK techniques.

|   |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t | data.win.system.opcode        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| t | data.win.system.processID     | 97464                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| t | data.win.system.providerGuid  | {5770385f-c22a-43e0-bf4c-06f5698ffbd9}                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| t | data.win.system.providerName  | Microsoft-Windows-Sysmon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| t | data.win.system.severityValue | INFORMATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| t | data.win.system.systemTime    | 2026-07-10T18:36:24.4634685Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| t | data.win.system.task          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| t | data.win.system.threadID      | 95836                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| t | data.win.system.version       | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| t | decoder.name                  | windows_eventchannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| t | full_log                      | { "win": { "system": { "providerName": "Microsoft-Windows-Sysmon", "providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}", "eventID": "1", "version": "5", "level": "4", "task": "1", "opcode": "0", "keywords": "0x0000000000000000", "systemTime": "2026-07-10T18:36:24.4634685Z", "eventRecordID": "145181", "processID": "97464", "threadID": "95836", "channel": "Microsoft-Windows-Sysmon/Operational", "computer": "Sarvesh", "severityValue": "INFORMATION", "message": "\nProcess Create:\n\nRuleName: -\n\nUtcTime: 2026-07-10 18:36:24.460\n\nProcessGuid: {5bad5dcd-3ba8-6a51-h7h4-a0a0a0a0a07a0}\n\nProcessId: 106680\n\nImage: C:\\Windows\\System32\\WindowsPowerSh" } } } |
| t | id                            | 1783708584.6266420                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| t | input.type                    | log                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| t | location                      | EventChannel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| t | manager.name                  | soc-lab                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| t | rule.description              | Sysmon: Suspicious PowerShell execution detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| # | rule.firedtimes               | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| t | rule.groups                   | local, sysmon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| t | rule.id                       | 100002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| # | rule.level                    | 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| t | rule.mail                     | true                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| t | rule.mitre.id                 | T1059.001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| t | rule.mitre.tactic             | Execution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| t | rule.mitre.technique          | PowerShell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| t | timestamp                     | Jul 10, 2026 @ 14:36:24.870                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Figure 9. PowerShell Attack Chain Observed Within the Wazuh Platform

Figure 9 illustrates the complete PowerShell attack chain as observed within the Wazuh platform. Correlating the generated telemetry with custom detection rules enabled the simulated attack sequence to be reconstructed and analyzed from an analyst's perspective.

## **Phase B: Living-off-the-Land (LOLBIN) Attack Chain**

### **Objective**

The second simulation evaluated the SOC's ability to detect the abuse of legitimate Windows utilities commonly leveraged by attackers to evade traditional security controls.

### **Attack Sequence**

The following LOLBIN techniques were executed during the simulation:

- Certutil
- Mshta
- Rundll32

Execution of each utility generated endpoint telemetry that was collected by Sysmon and processed by Wazuh. The corresponding detection rules successfully identified each LOLBIN technique, demonstrating the effectiveness of the custom detections in recognizing trusted Windows binaries used for malicious purposes.

The investigation confirmed that the generated telemetry provided sufficient context to reconstruct attacker activity and classify each technique using the MITRE ATT&CK framework.



*Figure 10. Living-off-the-Land Attack Chain Validation*

Figure 10 demonstrates the successful detection of multiple LOLBIN techniques executed during the simulation. The collected telemetry provided sufficient evidence to reconstruct the attack sequence and classify each technique using the MITRE ATT&CK framework.

## 6. Investigation Findings & Lessons Learned

The SOCForge project successfully demonstrated that a functional Security Operations Center can be developed using open-source security technologies while maintaining workflows similar to those found in enterprise environments. Throughout the project, endpoint telemetry generated by Sysmon was successfully collected, processed, and analyzed by the Wazuh platform, allowing simulated attacker activity to be detected and investigated.

The detection engineering process validated that custom detection rules were capable of identifying suspicious PowerShell execution, download activity, and LOLBIN techniques. Mapping each detection to the MITRE ATT&CK framework provided standardized context that improved investigation consistency and demonstrated how observed behaviors relate to recognized adversary tactics and techniques.

Several important lessons were learned during the development of the lab. High-quality endpoint telemetry is essential for effective threat detection, while carefully designed detection rules significantly improve the accuracy of security monitoring. Additionally, successful investigations depend not only on generating alerts but also on providing analysts with sufficient contextual information to reconstruct attacker activity and understand the progression of an attack.

Future enhancements to SOCForge will focus on expanding detection coverage, integrating additional threat intelligence sources, implementing automated response capabilities, and simulating more advanced attack techniques. These improvements will further increase the realism of the environment while continuing to develop practical skills in detection engineering, threat hunting, and incident response.

Overall, SOCForge demonstrates that a well-designed home laboratory can provide valuable hands-on experience with modern SOC operations, offering a practical platform for developing cybersecurity skills that closely reflect real-world security monitoring and incident investigation workflows.

## References

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